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Aside 1: de Bruijn and Good

A binary string is k -universal if it contains every binary k -mer once

Assign the k -mers to edges instead of nodes

Each node represents the overlap between adjacent edges

Glue together nodes with the same label

The corresponding graph is called the de Bruijn graph

Less repeats with big k

We want to find an Eulerian Path

Odd degree node $\Rightarrow$ No Eulerian Cycle

Eulerian Cycle is polynomial

Euler's Theorem for Directed Graphs

Balanced Graph: $\text{Indeg} = \text{Outdeg}$ for all nodes

Graph is Strongly Connected if you can go from any node to any other

Not balanced $\vee$ Not SC $\Rightarrow$ No Eulerian Cycle

Eulerian Cycle $\Leftrightarrow$ Balanced $\wedge$ Strongly Connected

PE1

Random walk starting on v_0

Must come back to v_0

Do it again on a unused edge (one of the vertices must be in the original cycle).

Then combine the cycles

For the genome problem just add a dummy edge between the two odd degrees.

So we've solved the problem (easy version)

Boosting Coverage through Read Breaking

Split each read into smaller pieces